

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

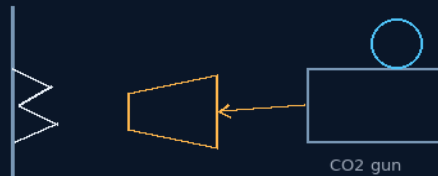
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 40

MAN-PORTABLE CABIN WALL-BREACH CO2 TAPERED PLUG REPAIR GUNS

No overhead repair units — the gun is the breach answer
Dress the hole — fire the plug — pressure seats it



dress the tear — drive the taper — pressure seats it

BAYONET DRESS — CO2 SHOT — JAG-LOCK RATCHET

14.7 PSI WEDGES THE PLUG TIGHTER EVERY SECOND

A leak, not a people-hoover — Phase 165 rides the wall plate

GP-IM-040

Revision v1.1 — 21 August 2026

41 of 290

VETTED: PASS CONCEPT — SEALING NUMBERS ON THE BENCH

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 40

PHASE 40: MAN-PORTABLE CABIN WALL-BREACH CO2 TAPERED PLUG REPAIR GUNS — STATUS:

CURRENT. The last line of defence on a cabin-wall breach: there are no overhead repair units — a puncture on a joining wall between two locked pods gets a crew answer, and the wall gun is it. A wall-recessed repair gun at every cabin — a hardened slide-out bayonet cone that forces torn sheet-metal teeth back into a uniform circular layout, then a CO2 dump-valve shot that drives a tapered silicone plug into the dressed hole; the jagged barbs bite the rubber wall, and the cabin's own 14.7 psi wedges the cone tighter with every second it holds. It seals a LEAK, not a people-hoover — Phase 165's operator notation rides on the wall plate.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-040, v1.1 — 21 August 2026. The forty-first manual of the program — 41 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the repair-gun law as seated (contents line, the four design bodies, the origin — verbatim) — §2 why it works (the seal physics — graded) — §3 the system on the wall — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

VERSION NOTE — v1.1, 21 AUGUST 2026 (SITTING 550): his reading-pass order, verbatim: 'remove all yellow there are no overhead repair units, and the bung physics was wrong, bungs in barrels try to push out not in'. EXECUTED: (1) six yellow passages struck exactly — the overhead-arms premise in the essence; 'Inner Joining' from the Core Objective; the overhead-arms clause from Operational Function; the robots-are-blind sentence in the origin's why-it-matters; the barrel-bung analogy in §2; the struck words in the §4 doctrine bullet. (2) Every overhead-repair-arm reference amended under the ruling — there are no overhead repair units; the gun is the breach answer in its own right (essence, §4, §7, §8). (3) §2 corrected: a barrel bung seats from the outside and pressure tries to push it OUT; this plug fires from the pressure side — the plug is the anti-bung. (4) His typed edit seated exactly as typed in §3:

'NB:there are several pug sizes'. The master keeps everything verbatim; the manual states the current machine.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 40: **Man-Portable Cabin Wall-Breach CO2 Tapered Plug Repair Guns**. Equips crew stations with lightweight, rapid-response sealant launchers. Fires tapered, quick-expanding composite plugs directly into emergency hull punctures to instantly seal atmospheric breaches using internal air pressure.'

THE CORE OBJECTIVE (yellow struck 21 August 2026 — 'Inner Joining' deleted on his kill order; the master keeps the verbatim): 'Core Objective: Localized Manual Air-Leak Containment on Walls'

WALL-MOUNTED TACTICAL ALLOCATION (VERBATIM)

- **Placement Grid:** Flush-mounted inside dedicated emergency recess fixtures on the interior cabin walls of every Phase 19/32 standardized pod.
- **Operational Function:** Manually deployed to seal bulkhead micro-breaches. (The overhead-arms clause struck 21 August 2026 — there are no overhead repair units; the gun covers for no machine.)
- **What the gun is really sealing:** a leak, not a people-hoover — Phase 165's operator notation applies. Speed still matters: every second of outflow is atmosphere the ship must replace, and the breathing clock is running for anyone sharing the compartment.

HARDENED EDGE-TEMPLATING BAYONET (VERBATIM)

- **Component Geometry:** Lower-rail mounted, slide-out linear mechanical bayonet.
- **Structural Interface:** Tipped with a high-hardness metal cone matrix driven directly into jagged, irregular metal punctures.
- **Action:** Plastically deforms and forces inward-facing torn sheet-metal teeth back outward, re-shaping the damage into a uniform circular layout optimized to match the repair plug diameter.

CO2 BALLISTIC PROJECTILE LAUNCH MATRIX (VERBATIM)

- **Propulsion Driver:** Upper-mounted, rapid-thread screw-on liquid CO2 pressure canister.
- **Actuation Valve:** High-velocity pneumatic dump-valve system. Launches the plug with a high-intensity gas shockwave to overpower the escaping cabin airflow.

JAG-LOCKING TAPERED SILICONE PROJECTILES (VERBATIM)

- **Ammunition Profile:** Steep, tapered cone-shaped projectiles manufactured from highly pliable, high-density silicone rubber compound.
- **The One-Way Ratchet Interlock:** Ballistic injection wedges the narrow nose into the cleared hole. The external jagged metal barbs of the tear bite directly into the expanding rubber wall, preventing the plug from blowing out under vacuum.
- **Atmospheric Self-Sealing Loop:** The 14.7 psi internal cabin barometric pressure acts on the wide flat rear face of the cone, wedging the tapered plug tighter into the metal collar to guarantee an immediate, airtight temporary seal until full module rebuild at a Phase 33 depot.

ORIGIN OF THOUGHT — PHASE 40 (VERBATIM)

Archer, 3:01 pm, 16 July: "A mobile repair gun, mounted on each cabin wall. It has an upper screw-on CO2 canister that is used to fire cone-shaped silicone rubber plugs into the walls when pod strikes occur — where two connecting pods prevent the overhead assembly from working on a joining wall face. The lower section has a slide-out bayonet with a metal cone tip, to give a basic edge clean to a closer plug shape. The internal jagged edge and air pressure should hold the plug seal in place temporarily."

Why it matters, in plain English: a sliding-cube translation takes time the air doesn't give you. So the last line of defense is a human with a wall-mounted gun: ram the bayonet to template the hole, fire a silicone cone into it, and let the cabin's own air pressure finish the seal. Two seconds, airtight, foolproof. It is the same philosophy as the Phase 39 sacrificial lock: soft material, intelligently placed, beats rigid precision in a crisis.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE BAYONET DRESSING, AFFIRMED: plastically deforming jagged puncture teeth back into a uniform circular layout with a hardness-tipped cone is straightforward mechanical repair — you cannot seal a shred, so you dress the wound first. The plug diameter matching the re-formed hole is the whole trick.

THE CO2 SHOT, AFFIRMED IN PRINCIPLE: a compressed-gas launcher accelerating a plug is $F = P \cdot A$ arithmetic. The honest caution the vet seats with the pass: the escaping atmosphere pushes the OTHER way — the launcher must overpower the outflow long enough for the plug nose to engage, and 'instantaneously seals' is a test result, not an assumption (§9).

THE ONE-WAY RATCHET, AFFIRMED: ballistic injection wedges the narrow nose; the tear's own jagged barbs bite into the expanding rubber wall against blow-out. The damage becomes part of the fastener — the irregularity that made the hole is what holds the plug.

THE SELF-SEATING LOOP, AFFIRMED — AND THE BUNG ANALOGY CORRECTED (his ruling, 21 August 2026: 'the bung physics was wrong, bungs in barrels try to push out not in'): a barrel bung seats from the OUTSIDE — internal pressure tries to push it out and only friction holds it. This plug is the anti-bung: fired from the pressure side, nose to vacuum, the cabin's 14.7 psi acts on the wide rear face and wedges the taper tighter the harder the leak pushes — pressure is the plug's ally, not its enemy — holding until a full module rebuild at a Phase 33 depot.

§3 — THE SYSTEM ON THE WALL

- THE ALLOCATION: flush-mounted in dedicated emergency recess fixtures on the interior cabin walls of every Phase 19/32 standardized pod — crew-reachable, no power required to grab.
- THE BAYONET: lower-rail slide-out linear mechanical bayonet tipped with a high-hardness metal cone matrix — driven into the puncture to force torn teeth outward into a uniform circular layout matching the plug diameter.
- THE DRIVER: upper rapid-thread screw-on liquid CO2 canister behind a high-velocity pneumatic dump valve — the gas shockwave launches the plug against the escaping airflow.
- THE AMMUNITION: steep tapered cone projectiles of highly pliable high-density silicone rubber — nose wedges, barbs bite, pressure seats. NB:there are several pug sizes
- THE SEAL: 14.7 psi cabin barometric pressure acts on the cone's wide flat rear face — an immediate airtight temporary seal until full module rebuild at a Phase 33 depot.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE CREW-ANSWER DOCTRINE (the ROBOTS-ARE-BLIND doctrine, struck and re-seated 21 August 2026): there are no overhead repair units — nothing was ever going to cover a wall between pods. What stands: a sliding-cube translation takes time the air doesn't give you, so the last line of defence is a person with a simple tool.
- THE LEAK-NOT-HOOVER NOTATION (Phase 165's operator ruling, carried on the wall plate): what the gun seals is a leak, not a people-hoover — Hollywood decompression is not the physics; the breathing clock is the real enemy, and every second of outflow is atmosphere the ship must replace.
- THE DAMAGE-IS-THE-FASTENER DOCTRINE (seated in the law): the jagged barbs of the tear bite the plug — the irregularity is used, not fought.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 11, SEATED — the verdict: ' PASS concept. Open: maximum hole size, cabin pressure, plug dimensions, CO₂ pressure and sealing time.' Mechanical hole preparation graded PASS, CO₂ launch graded PASS in principle ($F = P \cdot A$), with the outflow caution seated: the launcher must overcome the airflow long enough for the plug to engage. The vet's bench, quoted for the workshop wall: 'The required test is wonderfully simple: known hole size → known cabin pressure → plug mass → CO₂ pressure → launch velocity → leakage rate before/after.'

§6 — BUILD SEQUENCE

- STEP 1 — Recess the fixtures: the wall allocation grid across every Phase 19/32 pod cabin; gun, bayonet, canister and plug quiver in one grab.
- STEP 2 — Prove the bayonet: the hardness cone against representative torn-sheet coupons; the re-formed hole gauge-matched to the plug diameter.
- STEP 3 — Rate the driver: CO₂ canister pressure, dump-valve timing and plug muzzle velocity benched against worst-case outflow (§9).
- STEP 4 — Fire the qualification series: the vet's test — known hole, known pressure, known plug; leakage rate before and after, sealing time on the clock.
- STEP 5 — Drill the crew: dress, load, fire, verify — the two-minute breach drill until it is boring; the Phase 165 leak-not-hoover notation read at every drill.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASES 19/32 — the standardized pods (GP-IM-019, GP-IM-032 v1.2): the cabin walls the guns recess into.
- PHASES 6/8 — reference struck under his ruling, 21 August 2026: 'there are no overhead repair units'. The gun was never cover for a machine; it is the seated breach answer in its own right.
- PHASE 165 — the operator notation: a breach is a leak, not a people-hoover — seated on the wall plate.
- PHASE 33 — the outer-rim depots (GP-IM-033): where the full module rebuild happens after the temporary seal brings everyone home.

§8 — DO-NOT-BUILD

- NEVER fire into an undressed puncture — the bayonet re-forms the hole first; a plug into a shred is a leak with a plug in it (§1).

- NEVER assume 'instantaneous' — sealing time against outflow is a measured number, owed on the bench (§5, §9).
- NO waiting on a machine that does not exist — there are no overhead repair units (his ruling, 21 August 2026); the wall gun is the seated answer (§4).
- NEVER treat the seal as the repair — it is a pressure-tight temporary; the module rebuilds at depot (§1).

§9 — OWED

THE BENCH, OWED (the vet's open register, seated): maximum hole size, cabin pressure, plug dimensions, CO2 pressure and sealing time — the wonderfully simple test: known hole size, known cabin pressure, plug mass, CO2 pressure, launch velocity, leakage rate before/after. All owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-040, v1.1 — CURRENT, seated law, master v2.1 (numerical print order). The forty-first manual of the program — 41 of 290. Built 21 August 2026, third of the 'next 10' shift; revised same day on his reading pass (sitting 550).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on order (seated in sitting 537): 'ok next 10'. The phase's own chain, all seated in the master: the contents line and the design bodies (July truncations preserved as seated); the 12 August naming batch; the 15 August wording kills carried inline; the vet batch 11 grade in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete — and yellow-mark strikes in the manual delete exactly (the master keeps all). v1.1: six yellow strikes + two rulings + one typed edit, seated under 'remove all yellow there are no overhead repair units, and the bung physics was wrong, bungs in barrels try to push out not in'. Next version triggers: the §9 benches or his word.

END OF MANUAL — PHASE 40